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PROJECT AND TEAM INFORMATION

Project Title

AlgoVision: Interactive DSA Learning, Visualization and Practice Platform

Team Name: *Team #*	*AlgoVision*
Team member 1 (Team Lead) *(Last Name, name: student ID: email, picture):*	*Abhishek – 2510012494* *abhishek.in9900@gmail.com* 
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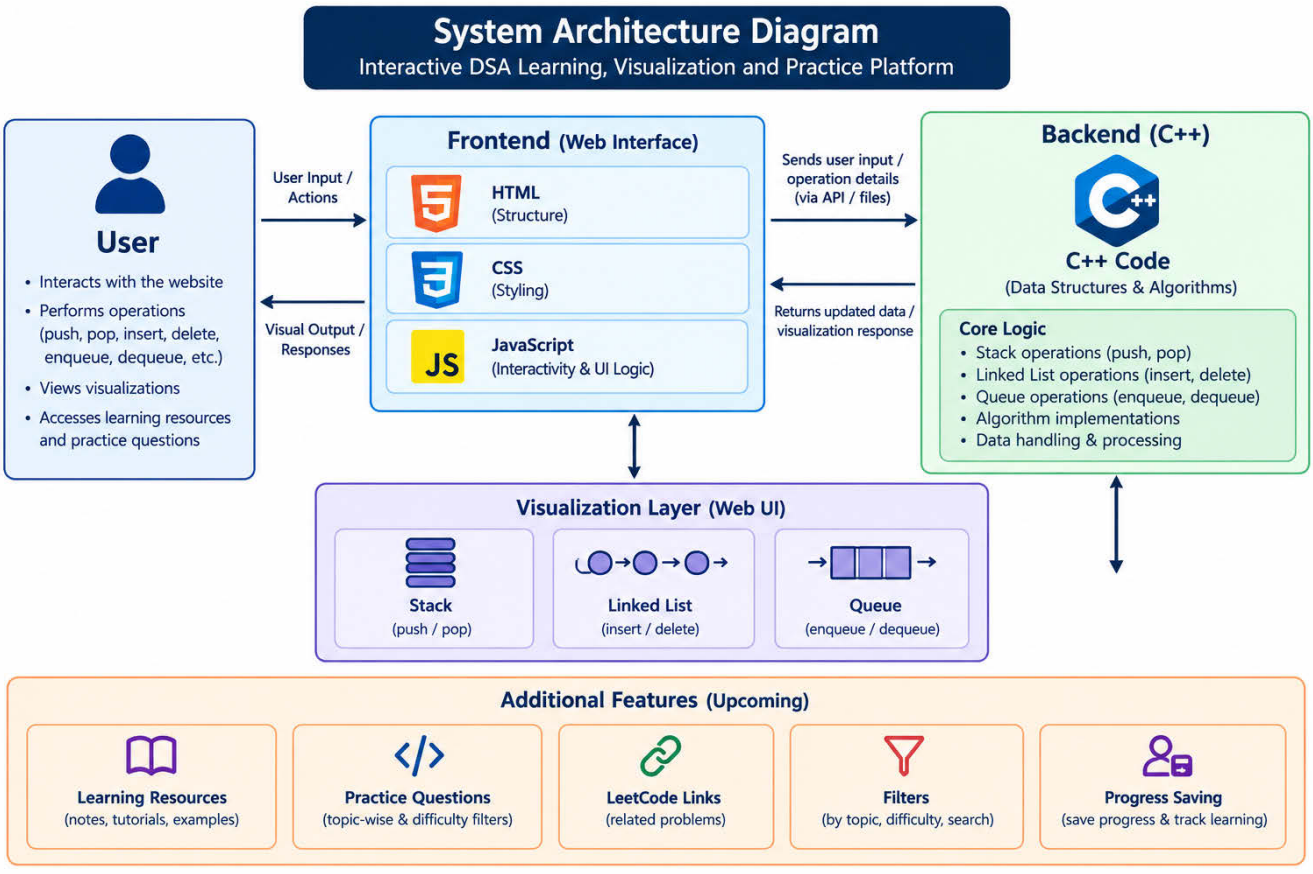
PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract (2 pts)

Any programmer who's studied DSA has probably experienced this: You read the code and still don't have a clear understanding of how the data is being manipulated. This is the gap we are trying to fill with AlgoVision. It's a project where users can observe data structures and algorithms at work. We currently have visualizers for Stack, Linked List, and Queue. Users can manipulate the structures by pushing, popping, inserting, and deleting elements and see the visual representation of the structures change. The website is developed by using HTML, CSS and JavaScript, and the data structures andalgorithms logic is programmed by using C++. We have a lot of additional features we want to add to the website. We want to add learning resources, practice questions, and links to related LeetCode problems.

Updated Project Approach and Architecture (2 pts)

In the beginning stages of AlgoVision, we identified key data structures that students learn through their DSA lessons. Currently, there are three different visualizers, namely Stack, Linked List and Queue. While the Stack section contains the push and pop functionality, the Linked List enables addition and deletion while the Queue section has enqueueing and dequeuing functionality. In technical terms, we built the front end using HTML and CSS, javascript is used for interaction and updating of website content while the backend is done using code. So what is next? The plan is to include content on different topics as well as various types of questions based on them. We'd also like to include filters for each topic that will allow users to filter based on search and difficulty questions. Another upcoming feature includes saving user progress.



Tasks Completed (7 pts)

Task Completed	Team Member
Worked on project planning and coordinating the initial development of AlgoVision.	Abhishek
Worked on the visualizer interface and prepared project documentation.	Chaitanya Maithani
Worked on testing the existing data structure operations and organizing the project data.	Arihant Rawat
Contributed to website development and helped organize and test the existing project modules.	Chaitanya Kharkwal

Challenges/Roadblocks (7 pts)

Of all our challenges while developing AlgoVision, the biggest one was achieving clear representation of information structure processes on the screen. pointer changes, It was easy to draw but it was really difficult to modify the visualization depending on the Drawing the visualizations was the easy part, but making them work correctly with the actual C++ code turned out to be much harder than we expected. The real challenge was keeping the web interface and the C++ implementation separate while making sure they both worked as intended. We had to be careful not to mix up their roles. information after each user move. In addition to technical challenges, there was also a 'problem' of social nature. The C+ code and web visualization play different roles and it proved to be quite difficult to keep them separate from each other without mixing them up. So far, our strategy has been really simple and clear.

Tasks Pending (7 pts)

Task Pending	Team Member (to complete the task)
Coordinate the further development activities and review the project status.	Abhishek
Develop website for main use and arrange learning material of DSA using C++.	Chaitanya Maithani
Add practice questions, topic-wise search, difficulty filters and LeetCode links.	Arihant Rawat
Implement step-by-step visualization controls, browser-based progress tracking and help test the modules.	Chaitanya Kharkwal

Project Outcome/Deliverables (2 pts)

AlgoVision started from an annoyance, while studying DSA we had a feeling that going through code never gave us the whole picture, we couldn't grasp what's actually happening inside a data structure, hence we decided to make what we wish we had at the time, a website where you perform an operation and see what happens (basically a visualizer). So far we have created visualizers for Stack, Linked list and Queue and implemented C++ code for the basic operations on them, but we want to develop this into a whole learning experience, adding the lesson content for all Data Structures and Algorithms topics and providing definitions in simple language and examples in C++. After learning a new concept, one can test their understanding with the practice questions we will provide that one can search, filter and relate to similar LeetCode questions. In the future we want to implement step-by-step controls, so one can go through the steps of an operation and inspect them, and track progress. Overall the idea is humble and simple, give DSA a face, make it understandable and relatable.

Progress Overview (2 pts)

AlgoVision started from an annoyance, while studying DSA we had a feeling that going through code never gave us the whole picture, we couldn't grasp what's actually happening inside a data structure, hence we decided to make what we wish we had at the time, a website where you perform an operation and see what happens (basically a visualizer). So far we have created visualizers for Stack, Linked list and Queue and implemented C++ code for the basic operations on them, but we want to develop this into a whole learning experience, adding the lesson content for all Data Structures and Algorithms topics and providing definitions in simple language and examples in C++. After learning a new concept, one can test their understanding with the practice questions we will provide that one can search, filter and relate to similar LeetCode questions.

Codebase Information (2 pts)

GitHub Repository:MoonRooter/AlgoVision-
The project code is maintained in a GitHub repository so that the team can access and update the files during development. The repository includes the web-based visualizer for Stack, Linked List and Queue along with the C++ program of data structures operations.
Branch: Main branch (verify on GitHub before submission).

Testing and Validation Status (2 pts)

Test Type	**Status (Pass/Fail)**	**Notes**
Stack Operations	Pass	Testing Push, Pop and Reset functions.
Operations on Linked Lists	Pass	Test for insertion and deletion
Queue Manipulations	Pass	Enqueue, Dequeue and Reset tested.
Visualizer Interface	Pass	Verified whether the screen is updated after operations.

Deliverables Progress (2 pts)

- Stack Visualizer: Done. Users can do push and pop operations and see how the structure is affected.
- Linked List Visualizer: Completed. Users can visualize insertion and deletion operations.
- Queue Visualizer: Completed. Users can perform enqueue and dequeue operations.
- C++ Implementation: The data structures are implemented using C++.
- DSA Learning Content and C++ Examples: Planned for further development.
- Practice Questions and Difficulty Filters: Planned for future updates.
- LeetCode Links and Search Feature: Planned for future updates.
- Step-by-Step Controls and Progress Tracking: Planned for future updates.